DAVID ARTHURIAN

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EXPERIENCE

Northrop Grumman **May 2025 - Present**

Electrical Engineer

- Serve as Cognizant Engineer for the Electronic Power Subsystem (EPS) on NASA's HALO spacecraft, leading design, integration, and verification of power distribution hardware supporting critical spaceflight operations.
- Develop, analyze, and validate power electronics and PCB-level hardware, ensuring high-reliability operation in radiation and thermal environments.
- Collaborate with cross-functional teams (systems, mechanical, software, and suppliers) to drive requirements, root-cause hardware issues, and deliver mission-critical hardware under aggressive schedules.
- Conduct circuit-level simulations (LTSpice, OrCAD, Simulink) and hardware bring-up/debug using oscilloscopes, power analyzers, and automated test scripts (Python/Matlab).
- Lead failure analysis and corrective actions for hardware anomalies, improving reliability and reducing integration risk.
- Author and review technical documentation, schematics, and test procedures to meet NASA and industry standards

Sempra Energy **June 2023 - May 2025**

Systems Engineer

- Collaborated with an engineering team to design and optimize Southern California CNG/LNG distribution systems for a large-scale critical infrastructure network serving over 500,000 residential, commercial, and industrial customers.
- Developed real-time monitoring and analytics tools using SCADA, GIS, SAP, and SQL to enhance system performance.
- Created emergency response dashboards for system engineers, ensuring rapid diagnostics and failure mitigation.
- Led instrumentation and control system projects, integrating hardware and software for high-reliability gas distribution.
- Conducted data modeling with SQL Data to implement system improvements to improve CNG distribution by 50%.
- Compiled data, created dashboards for emergency scenarios, and provided on-call support for region engineers.
- Designed and executed six-figure capital projects, achieving 25% cost savings and delivery ahead of schedule.

PROJECTS

Embedded Systems - Video Game **Jan. 2024 - May 2024**

- Designed and built an interactive, sensor-based multiplayer video game with IoT communication.
- Created hardware-based motion controllers using Arduinos, IMU sensors, and LEDs for gesture-based input.
- Embedded real-time embedded control and wireless communication protocols for multiplayer synchronization.
- Engineered the game logic in Python and developed a Unity-based graphical interface for stable and fun gameplay.

Micromouse Competition - 1st Place **April 2023 - May 2023**

- Programmed and assembled an autonomous robotic vehicle for maze navigation using PID controllers and IR sensors.
- Integrated real-time embedded control algorithms in C++, optimizing performance for speed and accuracy.
- Achieved 1st place out of 30 participants, completing the challenge with a record time of sub 10 seconds.

Network Attached Storage (NAS) **July 2022**

- Produced and built a custom NAS system using Raspberry Pi 4 and RAID-configured HDDs to store personal data.
- Configured network protocols and security layers for secure cloud data storage and remote access by individuals.

SKILLS

C++, Python (Aysncio, Matplotlib, Numpy, OpenCV , Pandas, SciPy), IMB DOORS Verilog, MATLAB, CREO, SQL, VHDL, Cameo, Altium, LTSpice, OrCAD, PCB Design, SCADA, SAP, GIS, Simulink, Git, Windchill, VSCode

EDUCATION

The University of California, Los Angeles **Sept. 2020 - June 2024**

B.S. Electrical Engineering, IEEE Member, HKN Society *GPA: 3.75*

EIT Certificate in Electrical Engineering #182187 **June 2024**